

DOCUMIND — Product Requirements Document (v1.0)

1. Product Overview

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DOCUMIND	
"An AI-powered document intelligence platform that helps credit and financial analysts quickly understand financial documents with grounded summaries and visual citations."	
Target Launch: 3 Phases → April 5 / April 25 / May 10	
Users:	10 analysts (MVP)
Documents:	100 docs, ~2MB avg
Stack:	Flask + Next.js + Free LLM Models

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2. Problem Statement

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TODAY (Without Documind)

- Analysts manually read 10–50 page financial PDFs
- No quick way to get grounded summaries of borrower health
- Tables, charts, invoices require manual interpretation
- Junior analysts lack guidance interpreting financial data
- No memory of what an analyst cares about across sessions
- Evidence gathering for credit decisions is slow and error-prone

TOMORROW (With Documind)

- Upload PDF → get accurate summary in seconds
- Every claim backed by clickable citation → exact PDF page
- AI educator explains terms and walks through data
- System learns analyst preferences over time
- Proactive alerts surface inconsistencies and spikes
- 50% reduction in document review time

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3. Target Users

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USER			AI SKILL			PRIMARY TASKS		
Credit Analysts			Beginner			Review borrower financials		
Operations			Beginner			Summarize company performance		
Financial Analysts			Beginner			Validate financial data		

Key implication: UI must be self-explanatory.
No AI jargon. No prompt engineering required.
The system does the thinking — the user just uploads and reads.

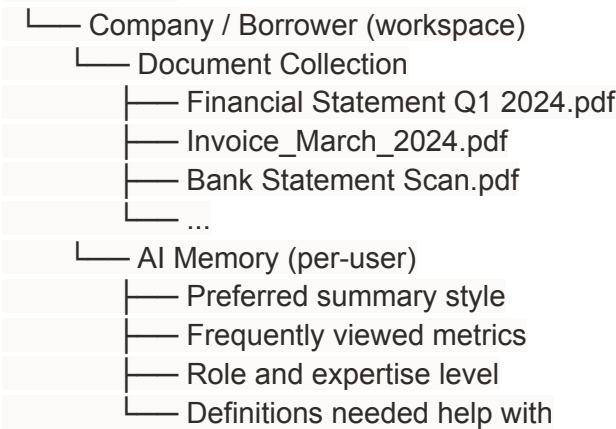
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4. Information Architecture

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WORKSPACE STRUCTURE

User Account



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5. Core Features — MVP

Feature Map

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DOCUMIND MVP FEATURES			
PHASE 1 — April 5th			
F1: Document Upload & Processing			
F2: LLM Summarization with Citations			
F3: Semantic Search (Top-K Retrieval)			
F4: Summary Dashboard (Proactive Insights)			
PHASE 2 — April 25th			
F5: Harvey-Style PDF Viewer (Exact Visual Render)			
F6: Citation Click → Navigate to PDF Page			
F7: Split-Pane Layout (Summary PDF Viewer)			
PHASE 3 — May 10th			
F8: AI Educator Agent			
F9: User Memory System			

Model	F10: Proactive Anomaly Detection
Model 1	~0.85
Model 2	~0.75
Model 3	~0.65
Model 4	~0.55
Model 5	~0.45
Model 6	~0.35
Model 7	~0.25
Model 8	~0.15
Model 9	~0.05
Model 10	~0.02

F1: Document Upload & Processing

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STATUS: Partially built (upload + embeddings exist)

REQUIREMENTS:

- Accept: PDF, Scanned PDF, Images
- Max size: 4MB per file
- Processing pipeline:
Upload → Detect type → Extract text/tables → Chunk → Embed → Store

EXTRACTION MUST CAPTURE:

Data Type	What to Extract
Text	Content + page number
Tables	Rows, columns, cell values + page
Numbers	EBITDA, cash flow, debt, totals
Invoice	Totals, payment terms, dates
Charts	Description (via vision model)

SCANNED PDF HANDLING:

- Auto-detect if page has extractable text
- If not → run OCR
- Flag OCR uncertainty to user

STORAGE:

- Chunks stored with: text, page_number, document_id, element_type
- Embeddings stored in existing vector DB – documents_block

ACCEPTANCE CRITERIA:

- User uploads PDF → processing completes within 30 seconds
- Extracted text is accurate for digital PDFs
- Scanned PDFs processed with OCR fallback
- All chunks linked to source page number
- OCR uncertainty flagged visually

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F2: LLM Summarization with Citations

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REQUIREMENTS:

- User uploads document → system auto-generates summary
- Default: Medium summary (half page, ~150–250 words)
- Option: Detailed summary (full page+) on user request
- Summary focuses on the entire document

CITATION RULES:

- Important claims MUST have citations [1], [2], etc.
- Each citation maps to a source page in the document
- Multiple evidence items per claim supported
- Example output:

DOCUMIND SUMMARY
Company XYZ — Q3 2024 Financial Review
EBITDA declined 8% to \$2.1M compared to prior quarter [1][2]. Cash flow from operations remained stable at \$890K [3]. Outstanding debt increased to \$5.4M due to new credit facility drawdown [1].
⚠ Unusual spike in invoice totals detected in March (\$340K vs \$120K avg) [4]
[1] Page 3 [2] Page 7 [3] Page 4 [4] Page 12

RETRIEVAL STRATEGY:

- Top-K retrieval (K=4-5 chunks)
- LLM interprets 3-4 candidate answers internally
- Displays ONLY the most confident answer
- If insufficient evidence → say "Insufficient evidence found in uploaded documents"

HARD RULES:

- Never answer without evidence from uploaded documents
- Never infer financial values not present in documents
- Flag OCR uncertainty explicitly
- Never use external knowledge for financial claims

ACCEPTANCE CRITERIA:

- Summary generated within 10 seconds of processing
- Every important claim has at least one citation
- Citations reference correct page numbers
- "Insufficient evidence" shown when data not found
- User can toggle medium ↔ detailed summary

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F3: Semantic Search

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REQUIREMENTS:

- Natural language search bar
- User types question → system retrieves relevant chunks
→ LLM generates grounded answer with citations
- Example queries:
 - "What is the company's current debt level?"
 - "Show me payment terms from the latest invoice"
 - "What was EBITDA last quarter?"

RETRIEVAL:

- Semantic search against vector embeddings
- Top-K (K=5) chunks retrieved
- LLM synthesizes answer from retrieved chunks
- Answer displayed as structured summary (not chat bubble)

ACCEPTANCE CRITERIA:

- ❑ Search returns results within 5 seconds
- ❑ Results are grounded in document content
- ❑ Each result shows citation to source page
- ❑ Irrelevant queries return "No relevant information found"

F4: Summary Dashboard (Proactive Insights)

REQUIREMENTS:

- Homepage acts as a proactive feed — NOT a blank search page
- After documents are processed, dashboard auto-populates:

DOCUMIND — Company XYZ Workspace

EBITDA

\$2.1M (-8%)

[View Source]

Cash Flow

\$890K (stable)

[View Source]

Debt

\$5.4M (+12%)

[View Source]

Invoices

\$340K March

[View Source]

ALERTS

Invoice spike: March \$340K vs \$120K avg

Inconsistency: Debt figure differs p3 vs p7

Ask Documind...

PROACTIVE INSIGHTS:

- Inconsistencies across document pages

- Unusual spikes in financial values
- Key metric highlights (EBITDA, cash flow, debt)

ACCEPTANCE CRITERIA:

- Dashboard populates automatically after document processing
- Insight cards show key metrics with source citations
- Alerts highlight inconsistencies and anomalies
- Each card/alert is clickable → opens source in viewer

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F5: Harvey-Style PDF Viewer

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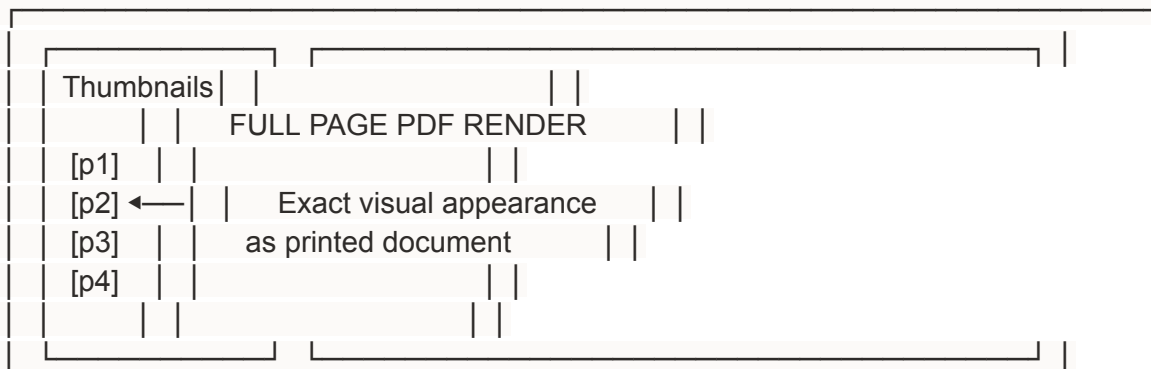
REQUIREMENTS:

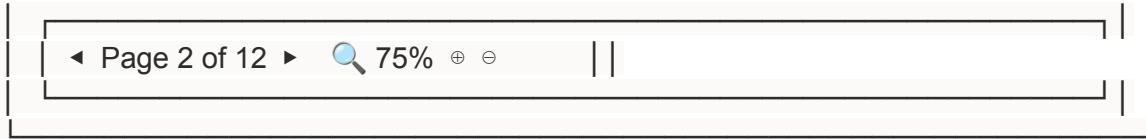
- Render uploaded PDF EXACTLY as original document
- Preserve:
 - Original layout
 - Original fonts and spacing
 - Scanned document appearance
 - Tables render exactly as in PDF
 - Charts render exactly as in PDF
- NOT an HTML conversion — actual PDF rendering

TECHNICAL APPROACH:

- PDF.js (Mozilla) for canvas-based rendering
- Page-by-page rendering with virtualization
- Zoom controls (fit-width, fit-page, manual zoom)
- Page navigation (scroll, thumbnail sidebar, page input)

VIEWER LAYOUT:





ACCEPTANCE CRITERIA:

- ☐ PDF renders identically to original document
- ☐ Fonts, tables, charts, images preserved
- ☐ Scanned PDFs render as original scanned images
- ☐ Smooth scroll and zoom
- ☐ Pages virtualized (no memory issues for large docs)
- ☐ Loads within 2 seconds for 2MB PDF

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F6: Citation Click → PDF Page Navigation

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REQUIREMENTS:

- User clicks citation [1] in summary
- PDF viewer navigates to the cited page
- Full-page preview of that page displayed
- Smooth scroll animation

INTERACTION FLOW:

1. User reads summary
2. Sees citation [1] next to a claim
3. Clicks [1]
4. Right panel (PDF viewer) navigates to page 3
5. Page 3 renders in full with original appearance
6. User verifies the claim against source

ACCEPTANCE CRITERIA:

- ☐ Clicking citation navigates to correct page
- ☐ Navigation is smooth (scroll animation)
- ☐ Correct page highlighted in thumbnail sidebar
- ☐ Works for multiple citations per answer
- ☐ Back button returns to previous view position

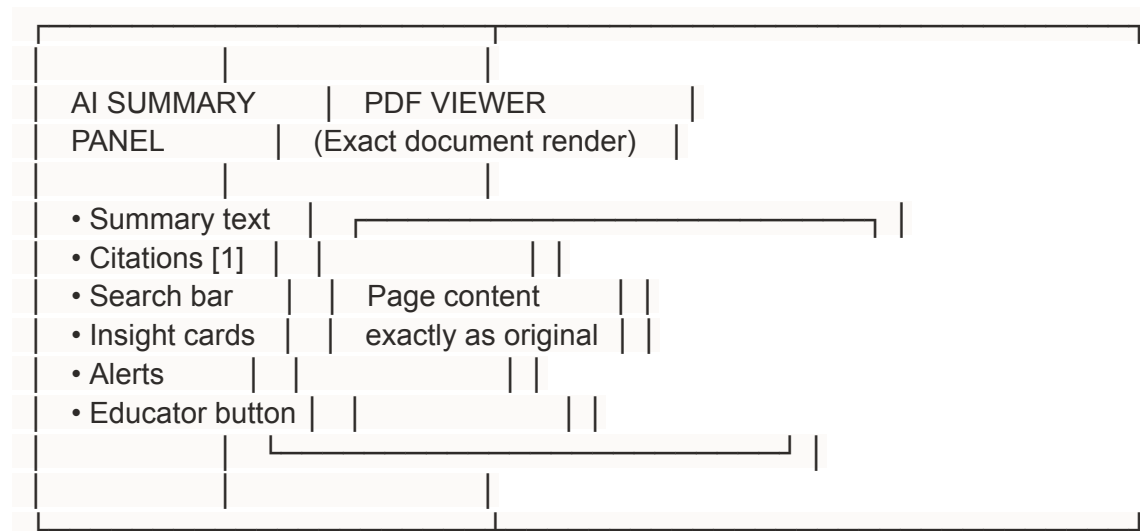
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F7: Split-Pane Layout

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REQUIREMENTS:

- Two-panel layout:
 - LEFT: AI Summary + Search + Insights
 - RIGHT: PDF Viewer (Harvey-style)



- Panels are resizable (drag divider)
- Panels are collapsible
- Default split: 40% left / 60% right

ACCEPTANCE CRITERIA:

- ☐ Split pane renders correctly
- ☐ Panels resizable via drag
- ☐ Citation click on left → navigates viewer on right
- ☐ Responsive on standard desktop screens (1280px+)

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F8: AI Educator Agent

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REQUIREMENTS:

- Educator is an AI agent that helps users understand financial data, terms, and document content
- Available on every summary/insight via "Explain" button
- Uses ONLY uploaded document content (no external knowledge)

CAPABILITIES:

- Explain finance terms simply ("What does EBITDA mean in this context?")
- Teach junior analysts how to interpret statements
- Walk through charts and tables step by step
- Answer follow-up questions about the summary
- Suggest next questions the user should ask

SELF-LEARNING MECHANISM:

- Educator maintains a markdown (.md) file per user
- After every conversation, the agent updates the file:
 - What the user asked
 - What the agent explained
 - What the agent got wrong (if corrected)
 - What the user struggled with
- This file is used to improve future responses

EXAMPLE educator_log_user_123.md:

# Educator Log — Analyst: John D.	
## Session: 2025-04-08	
- User asked: "What does EBITDA margin mean?"	
- Explained: EBITDA as % of revenue	
- User understood: ✓	
## Session: 2025-04-09	
- User asked: "Why is cash flow negative?"	
- Explained: Operating vs investing cash flow	
- User follow-up: "What's capex?"	
- Correction: User pointed out I missed a line item on page 4 — noted for future.	
## Patterns	

- User frequently asks about cash flow terms	
- Prefers short bullet explanations	
- Needs help with ratio interpretation	

INTERACTION MODEL:

- Button on every insight/summary: "🎓 Explain this"
- Opens educator panel (slide-in or modal)
- Conversational within that panel
- Grounded only in document content

ACCEPTANCE CRITERIA:

- ❑ Educator explains terms in simple language
- ❑ Educator only uses uploaded document content
- ❑ MD file created per user and updated per session
- ❑ User corrections logged and influence future answers
- ❑ Suggest 2-3 follow-up questions after each explanation

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F9: User Memory System

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REQUIREMENTS:

- System remembers per-user preferences:

MEMORY ITEM	EXAMPLE	
Summary style	"Prefers bullet points"	
Frequently viewed metrics	"Always checks EBITDA and cash flow first"	
Role and expertise	"Junior credit analyst"	
Definitions needed	"Needed help with leverage ratio, capex, DSO"	
Preferred output format	"Likes medium summaries with key metrics bolded"	

STORAGE:

- Per-user memory profile (JSON or structured store)
- Updated automatically based on usage patterns

- Also updated from educator agent interactions

USER CONTROL:

- Users can view what the system remembers
- Users can edit preferences manually
- Users can delete specific memory items
- Settings page: "What Documind knows about you"

HOW MEMORY IS USED:

- Summary generation uses memory to personalize output (e.g., always highlight EBITDA if user checks it often)
- Educator adapts explanation depth to expertise level
- Dashboard prioritizes metrics the user cares about

ACCEPTANCE CRITERIA:

- ☐ Memory profile created on first use
- ☐ Memory updates after each session
- ☐ User can view/edit/delete memory items
- ☐ Summaries adapt based on stored preferences
- ☐ No compliance restrictions (confirmed)

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F10: Proactive Anomaly Detection

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REQUIREMENTS:

- System automatically scans processed documents for:
 - Inconsistencies across pages (e.g., revenue on page 3 $\neq$ revenue on page 7)
 - Unusual spikes in values (e.g., invoice total 3x higher than document average)
- Alerts displayed on dashboard without user asking

ALERT FORMAT:

 Inconsistency Detected	
Revenue reported as \$4.2M on page 3 but	
\$3.9M on page 7.	

[View Page 3]	[View Page 7]

 Unusual Spike Detected	
March invoice total (\$340K) is 183% above the document average (\$120K).	
[View Source]	

ACCEPTANCE CRITERIA:

- ☐ Inconsistencies detected across pages automatically
- ☐ Unusual value spikes flagged with percentage context
- ☐ Alerts link to source pages in PDF viewer
- ☐ Alerts appear on dashboard without user query

6. Explicitly OUT of MVP Scope

FEATURE	WHEN	
Multi-document comparison	Post-MVP	
Team/shared memory	Post-MVP	
Agent workflows (automated)	Post-MVP	
External integrations (Drive, SharePoint, ERP)	Post-MVP	
Multi-language support	Post-MVP	
Excel/CSV ingestion	Post-MVP	
Sub-page-level citations (bounding box highlights)	Post-MVP	
Chart data extraction (structured, not description)	Post-MVP	
Dark mode	Post-MVP	
Mobile responsive	Post-MVP	
SSO/SAML	Post-MVP	

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7. UX Principles

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PRINCIPLE 1: PROACTIVE, NOT REACTIVE

- System surfaces insights before user asks
- Homepage is a feed, not a blank search box
- "The user should not have to think"

PRINCIPLE 2: GROUNDED TRUST

- Every important claim has a citation
- One click verifies against original document
- "Insufficient evidence" when uncertain
- Never fabricate financial data

PRINCIPLE 3: STRIPE-LEVEL CRAFT

- Clean, minimal, premium, calm, analytical
- No clutter, no unnecessary elements
- Typography-driven hierarchy
- Generous whitespace
- Micro-interactions that feel polished

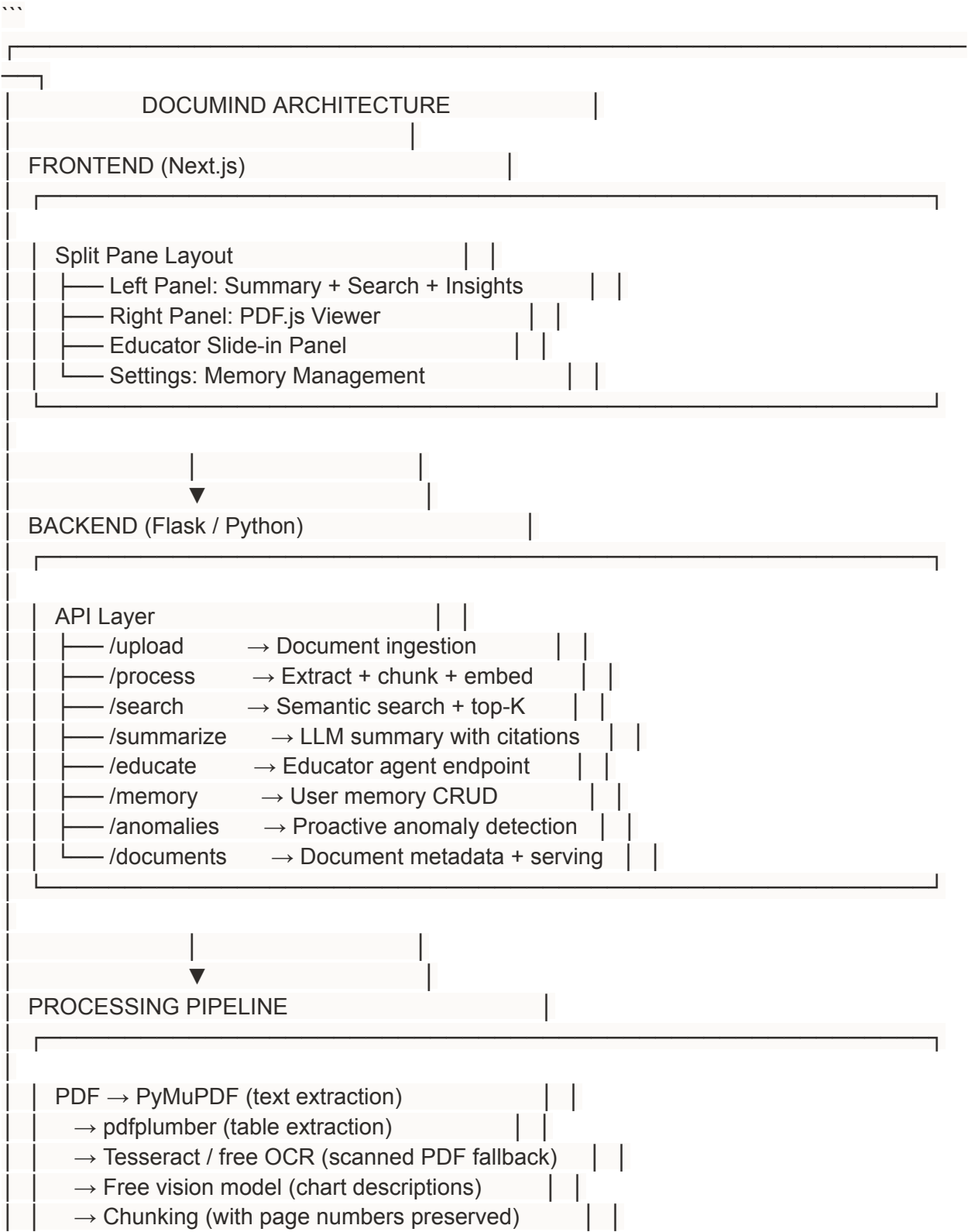
PRINCIPLE 4: BEGINNER-FRIENDLY

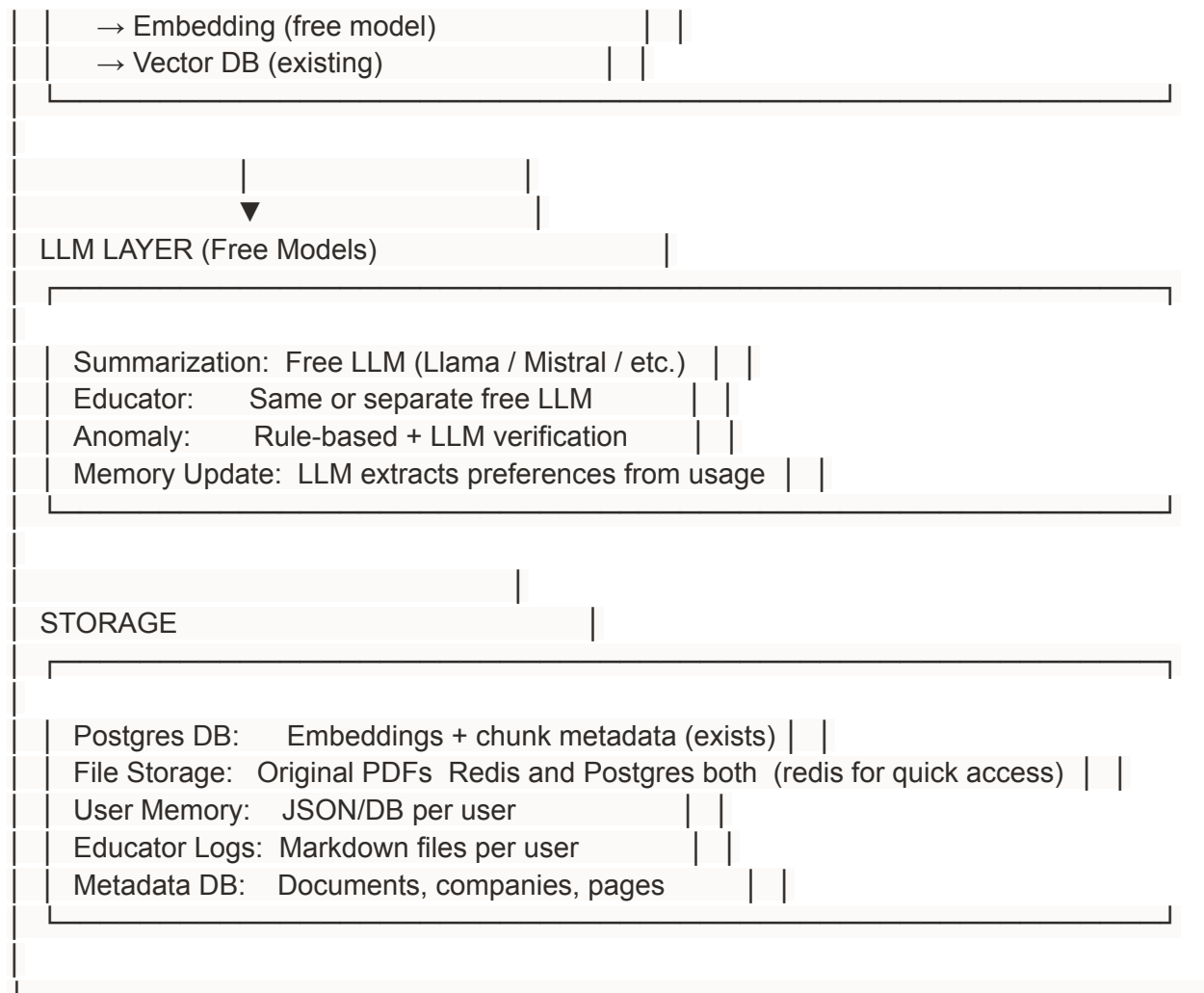
- No AI jargon in the interface
- Educator always available to explain
- System adapts to user's expertise level
- Progressive disclosure — simple first, detail on demand

PRINCIPLE 5: MEMORY-DRIVEN PERSONALIZATION

- System gets smarter per user over time
 - Summaries adapt to what the user cares about
 - Educator remembers what was taught before
 - User always in control of their data
- ...
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8. Technical Architecture





9. Data Model

```

```sql

-- DOCUMENTS
document {
 id UUID PRIMARY KEY
 company_id UUID → company.id
 filename TEXT
 file_path TEXT -- for PDF viewer serving

```

```

file_size INTEGER
page_count INTEGER
doc_type ENUM(pdf, scanned_pdf, image)
processing_status ENUM(uploading, processing, ready, failed)
ocr_used BOOLEAN
ocr_confidence FLOAT -- flag if low
uploaded_by UUID → user.id
uploaded_at TIMESTAMP
}

```

-- CHUNKS (for RAG)

```

chunk {
 id UUID PRIMARY KEY
 document_id UUID → document.id
 page_number INTEGER -- CRITICAL for citations
 content TEXT
 element_type ENUM(text, table, chart_description, invoice)
 embedding VECTOR
 created_at TIMESTAMP
}

```

-- SUMMARIES

```

summary {
 id UUID PRIMARY KEY
 document_id UUID → document.id
 summary_text TEXT -- with [1][2] citation markers
 summary_length ENUM(medium, detailed)
 citations JSON -- [{ref: 1, page: 3}, ...]
 generated_at TIMESTAMP
}

```

-- ANOMALIES

```

anomaly {
 id UUID PRIMARY KEY
 document_id UUID → document.id
 anomaly_type ENUM(inconsistency, spike)
 description TEXT
 pages_involved JSON -- [3, 7]
 severity ENUM(info, warning, critical)
 detected_at TIMESTAMP
}

```

-- USER MEMORY

```

user_memory {

```

```

id UUID PRIMARY KEY
user_id UUID → user.id
memory_key TEXT -- e.g., "preferred_metrics"
memory_value TEXT -- e.g., "EBITDA, cash flow"
source ENUM(auto, manual, educator)
updated_at TIMESTAMP
}

```

```
-- EDUCATOR LOGS
```

```

educator_session {
 id UUID PRIMARY KEY
 user_id UUID → user.id
 document_id UUID → document.id
 question TEXT
 explanation TEXT
 user_correction TEXT NULL
 follow_ups JSON
 session_at TIMESTAMP
}

```

```
...
```

```

```

## 10. API Specification

```
...
```

ENDPOINT	METHOD	DESCRIPTION
/api/companies	GET	List user's companies
/api/companies	POST	Create company/workspace
/api/documents	GET	List documents in company
/api/documents/upload	POST	Upload document + trigger processing
/api/documents/:id	GET	Get document metadata
/api/documents/:id/file	GET	Serve original PDF file (for viewer)
/api/documents/:id/summary	GET	Get AI summary with citations
/api/documents/:id/summary	POST	Generate summary (medium/detailed)
/api/search	POST	Semantic search query
/api/anomalies/:doc_id	GET	Get detected anomalies
/api/educate	POST	Ask educator a question
/api/memory	GET	Get user's memory profile
/api/memory	PUT	Update memory item
/api/memory/:id	DELETE	Delete memory item

## 11. Delivery Timeline

### PHASE 1: LLM MODEL INTEGRATION

TARGET: APRIL 5

Week 1 (Mar 24–30):

- Enhance document processing pipeline
  - Add page number tracking to chunks
  - Add table extraction (pdfplumber)
  - Add OCR fallback for scanned PDFs
- Build /summarize endpoint
  - Top-K retrieval → LLM summary
  - Citation generation (page-level)
  - Medium vs detailed toggle

Week 2 (Mar 31–Apr 5):

- Build summary display UI (left panel)
  - Summary cards with citation markers
  - Search bar with natural language query
- Build dashboard with proactive insights
  - Key metric cards (EBITDA, cash flow, debt)
  - Anomaly alerts (inconsistencies, spikes)
- Integration testing with real financial PDFs

DELIVERABLE: User uploads PDF → gets summary with page citations

### PHASE 2: HARVEY-STYLE PDF VIEWER

TARGET: APRIL 25

Week 3 (Apr 7–13):

- Integrate PDF.js into Next.js
  - Canvas-based rendering
  - Page navigation + zoom
  - Thumbnail sidebar

- Serve original PDF from backend
- Handle rendering edge cases

Week 4 (Apr 14–20):

- Build split-pane layout
  - Left: summary panel
  - Right: PDF viewer
  - Resizable divider
- Implement citation click → page navigation
  - Click [1] → viewer scrolls to page 3
  - Smooth animation
  - Page highlight in thumbnails

Week 5 (Apr 21–25):

- Stripe-level UI polish
  - Typography, spacing, colors
  - Loading states and skeletons
  - Micro-interactions
- End-to-end testing: summary → citation → PDF page

DELIVERABLE: Full split-pane with exact PDF viewer + citations

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PHASE 3: AGENT INTEGRATION

TARGET: MAY 10

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Week 6 (Apr 28–May 4):

- Build educator agent
  - /educate endpoint
  - "Explain this" button on every insight
  - Slide-in educator panel
  - Follow-up question suggestions
- Build educator markdown logging system
  - Create MD file per user
  - Update after each conversation
  - Log corrections and patterns

Week 7 (May 5–10):

- Build user memory system
  - Auto-detect user preferences
  - Memory profile settings page
  - View/edit/delete memory items

- Personalize summaries using memory
- Full integration testing
- Bug fixes and polish
- Demo preparation

DELIVERABLE: Full platform with educator + memory + proactive insights

## 12. Risks and Mitigations

RISK	SEVERITY	MITIGATION
Free LLM quality insufficient for financial summaries	HIGH	Test multiple free models (Llama 3, Mistral, Phi-3) Fallback: low-cost API tier
OCR accuracy poor for scanned financial documents	HIGH	Use best free OCR (Tesseract 5 + preprocessing) Flag low-confidence regions
Table extraction fails on complex financial statements	MEDIUM	Multiple extractors (pdfplumber + camelot) Vision model fallback
PDF.js rendering issues with certain PDF formats	MEDIUM	Test with 20+ real PDFs early in Phase 2
Citation page numbers wrong	HIGH	Verify page tracking in chunking pipeline Add page validation step

Timeline too aggressive (7 weeks for all 3 phases)	MEDIUM	Phase 1 is non-negotiable Phase 2-3 can slip 1 week AI tools accelerate 30-40%
Memory system creates bias in AI responses	LOW	Always show evidence first Memory only affects format and presentation, not facts
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## 13. Success Metrics

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PRIMARY GOAL: Reduce document review time by 50%

KPI	TARGET	HOW MEASURED
Time to first insight (upload → summary displayed)	< 30 sec	Upload → summary time
Citation click-through rate	> 40%	% of users who click citations
Summary acceptance rate (user does not re-query)	> 70%	% of summaries accepted
Retrieval precision (correct page cited)	> 85%	Cited pages are correct



Docs processed per user/week	> 10	Usage
	tracking	

TOOL	ASSIGNED ROLE IN DOCUMIND BUILD
Claude Opus (Antigravity)	Architecture decisions Pipeline design Prompt engineering for summary + educator Data model review Complex debugging
Codex	Code generation and execution API endpoints React components PDF.js integration Database setup
Gemini	Chart/image understanding in documents Multimodal fallback for failed extraction Research alternatives when blocked
Claude	Code review Performance optimization Security review

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## ## 15. Approval

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Document Version: 1.0

Created: March 2025

Author: James Nazareth

Status: DEVELOPMENT WIP

Phase 1 Start: Immediately

Phase 1 Deadline: April 5, 2025

Phase 2 Deadline: April 25, 2025

Phase 3 Deadline: May 10, 2025